

RESEARCH ARTICLE

Predicting air-entraining in cement paste from the molecular attributes of nonionic surfactants with a multilayer method

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Abstract

The novelty of this study is to present a multilayer framework for predicting the air-entrained porosity of cement paste based on the molecular characteristics of nonionic surfactants. Air-entraining agents enhance concrete durability against freeze–thaw damage; however, their development is labor-intensive and cost-prohibitive. This research implements a multilayer approach by incorporating three hierarchical layers: the molecular properties of nonionic surfactants (Layer 1), their physicochemical characteristics (Layer 2), and the air-entrained microstructural porosity of hardened cement paste (Layer 3). By integrating key molecular parameters—such as hydrocarbon chain length, hydrophobicity, and molecular weight—this model effectively predicts the air-entrained porosity of cement paste. An extensive experimental study was conducted to characterize the physicochemical and microstructural properties of 59 distinct nonionic surfactants. To the best of our knowledge, this represents the first comprehensive dataset of molecular and physicochemical properties of air-entraining agents reported in the literature. Moreover, no prior study has established such a detailed link between the molecular characteristics of nonionic surfactants and cement microstructure. This dataset served as the foundation for developing the predictive model, which demonstrated the feasibility of this approach in predicting the air-entraining performance of nonionic admixtures. The developed model facilitates the rapid screening of candidate surfactants and the optimization of their molecular structure while minimizing the need for extensive experimentation. Furthermore, distinct trends emerged from the dataset, offering new insights into the interdependent properties that govern air entrainment in cementitious materials.

KEYWORDS

admixture, air-entraining, cement, machine learning, surfactants

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1 | INTRODUCTION

Admixtures are an essential part of concrete mix design and enable control over properties, including workability, setting time, strength, and durability by modifying the chemical properties and reactions in concrete.^{1,2} As such, admixtures are an important component of optimizing concrete properties and meeting structural and environmental requirements as the construction industry moves toward more sustainable and resilient materials.³ Freeze–thaw damage in concrete is a critical durability concern in wet and cold climates, where concrete structures are exposed to precipitation and cycles of freezing and thawing.⁴ When water penetrates the concrete pores and freezes, it expands in volume and creates internal hydraulic pressure that can cause cracking and structural degradation over time.^{4,5} This progressive degradation compromises the mechanical integrity of concrete, necessitating frequent repairs and posing structural safety risks. To enhance freeze–thaw resistance, air-entraining admixtures (AEAs) are incorporated into concrete mixtures, facilitating the formation of a stable, finely dispersed air void system,^{6,7} commonly referred to as the air-entrained voids. Air-entrained voids provide a release mechanism for the hydraulic pressures caused by ice formation and expansion and, in turn, increase the concrete's freeze–thaw resistance.^{6,8} AEAs are generally used to generate a total air-entrained porosity of 5%–6% in concrete, void sizes ranging from 10 to 1000 μm ^{1,5} and a typical spacing factor of 200 μm per ACI 201⁹ to effectively inhibit the freeze–thaw damage. In this paper, porosity only refers to the total volume fraction of air-entrained voids in the range of 10–1000 μm and does not include other porosities such as capillary porosity.

Many AEAs belong to the surfactant chemical family, which are amphipathic molecules with a hydrophilic head and a long hydrophobic tail.^{6,8,10} Surfactants tend to accumulate at the interfaces (solid–liquid or liquid–air) and reduce the surface tension of the liquid.⁵ Although many surfactants have an ionic head, others, known as nonionic surfactants, do not have a charged head group. The behavior of AEAs depends on their molecular characteristics, which can influence the size, distribution, and stability of the entrained air bubbles.¹¹ Uncovering these relationships at a molecular level can lead to the development of more efficient AEAs and positively impact the durability of concrete in freeze–thaw cycles.

Traditionally, new admixtures are developed by incrementally modifying the structure of existing admixtures to improve their performance.¹² This approach is in line with classical paradigms of empirical and theoretical science.¹³ For instance, the molecular structure and prop-

erties of polycarboxylate ether superplasticizers are tuned experimentally by adjusting the ratio of methacrylic acid to poly(ethylene glycol) methacrylate or the length of polyethylene glycol chain.¹⁴ Likewise, for lignosulfonate-based superplasticizers, the adjustment happens in the ratio of sulfonate to alkyl.¹⁴ This labor-intensive process is costly and limits the discovery of novel chemical structures, particularly given the need to ensure compatibility with various binders and admixtures.^{14,15}

On the other hand, machine learning is a new paradigm where patterns in large amounts of data can be inferred using statistical methods.¹⁶ However, the acquisition of a large amount of experimental data for the development of machine learning models is a significant challenge.^{13,16} Recent advances facilitated the use of smaller datasets with machine learning approaches in materials science and engineering disciplines,^{12,14–21} which enable the prediction of new and optimized chemical structures in a fraction of time and cost. Furthermore, critical parameters can be identified more easily, and different chemicals can be screened in terms of their efficiency.^{13,16} Menon et al.¹⁴ applied a hierarchical machine learning methodology to develop models capable of predicting the workability of lignosulfonate, polynaphthalene, and polycarboxylate ether polymers in a cement paste system containing metakaolin. In another study, Menon et al.²⁰ used features representing functional groups and molecular architectures to elucidate the rheological properties of magnesium oxide slurries. The same group used a similar approach to predict 3D printability of silicon elastomers and alginate hydrogels^{19,17} and stress–strain properties of polymer films.¹⁸ Kayello et al.¹² and Shlonimskaya et al.¹⁵ employed features, including shape and length descriptors of amine and glycol ethers, to predict the rate of surface tension reduction and subsequently the performance of chemicals as shrinkage-reducing admixtures in ordinary Portland cement. Childs et al.²¹ developed a model capable of predicting the retarding performance of chemical admixtures in calcium sulfoaluminate cement using functional groups such as sulfonates and phosphonates as features and with a dataset of 23 molecules. Weis and Visco²² evaluated the environmental impact of chemical products and green compatibility of solvents—according to the chemical selection guide developed by Curzons et al.²³—by using the atomic descriptors with the signature method. Model-based approaches to simulate physicochemical interactions will enable researchers to speed up the chemical discovery process significantly and design across many permutations of molecular shapes and arrangements.¹² The model-based approach also helps researchers perform high-throughput screening of the material databases for the specific chemicals that possess the desired variables.¹³

Research on applying machine learning to optimize AEAs in cementitious materials remains limited, with fundamental mechanisms still not entirely understood. Thus, the novelty of this paper is to leverage recent advances in machine learning in materials science to develop a multilayer machine learning approach to predict the air-entrained porosity of the cementitious microstructure from the molecular properties of the nonionic surfactants. Nonionic surfactants were selected in this study to minimize electrostatic interactions with cement particles, thereby isolating and simplifying the effects of molecular structure on air entrainment. This choice allowed us to reduce system complexity and improve interpretability of the results, which was particularly important given the limited dataset size. The multilayer approach bridges the gap between molecular characteristics of nonionic surfactants and the microstructure of hardened cement paste by implementing a middle layer that contains information on the physicochemical properties of the nonionic surfactants. In other words, molecular properties are used to predict the physicochemical properties relevant to air-entraining, and then these physicochemical properties are utilized to predict the air-entrained porosity of cement paste. By systematically linking molecular attributes of nonionic surfactants to the microstructural characteristics of hardened cement paste, this study introduces a novel approach that has not been previously explored. This approach not only enables rapid screening of potential admixtures but also helps in a deeper understanding of the underlying mechanisms governing air entrainment in nonionic surfactants. The developed models were able to predict the air-entrained porosity of never-before-seen nonionic surfactants with a good fit and small error. Furthermore, influential features of the model were examined to elucidate the relationship among parameters that influence air entrainment in cement paste and gain new insights. Ultimately, this work advances the field by enabling a data-driven methodology for tailoring surfactant chemistry to achieve optimal air entrainment, a critical factor in enhancing the longevity and performance of concrete in freeze-thaw environments.

2 | MACHINE LEARNING METHODOLOGY AND SCHEME

The machine learning methodology of this study follows the multilayer approach in its layer arrangement to represent the entire spectrum of structure–property–performance.^{14,16,20} This is shown in Figure 1. The model is composed of three layers, which comprise information at the nano, micro, and meso scales, respectively. Each property of Layer 2 is predicted with a separate model from a

combination of relevant properties in Layer 1. Then the obtained properties of Layer 2 are used to predict the air-entrained porosity in hardened cement paste. The layers embody the concept of structure–property–performance by establishing the relationship between the molecular properties of nonionic surfactants, the resulting physicochemical properties, and the air-entrained porosity in the hardened paste.

Layer 1 consists of the molecular properties, which are obtained directly from the manufacturer and used with little modification. These features are listed in Table 1 as the molecular properties and include properties such as the molecular weight (Mw), length of the hydrocarbon chain, and hydrophobicity of the molecule. However, not all the properties were used in Layer 1 simultaneously, and a feature selection algorithm was used to select the influential properties, which are later discussed in Section 4.1.4. It was important to select properties that are commonly used in nonionic surfactants' catalogues. This facilitates the use of the model in predicting preferable air-entraining surfactants from the already available chemicals, without the need for extensive laboratory experimentation.

Layer 2 is composed of the physicochemical properties of the surfactants in the pore solution that are involved in the air-entraining mechanism of nonionic surfactants in cement paste. It was important to choose the properties that are shown to influence the air-entrained void microstructure. Based on previous studies, surface tension, foam capacity and foam stability at 30 min, and particle hydrophobization have been selected.⁵

Layer 3 is the output layer and contains the air-entrained porosity of the hardened cement paste microstructure. Air-entrained porosity is obtained by numerical analysis on the 3D reconstructed model of the micro-computed tomography scans of the hardened cement paste samples.

Several considerations were accounted for during data collection and model development. Charge is a critical property of ionic polymers that influences multiple aspects of air entrainment, including adsorption, bubble interfacial viscoelasticity, and foam stability.^{5,11,24} However, because the surfactants analyzed in this study are nonionic, charge was excluded from the air-entraining interactions. According to Liu et al.²⁵ the hemi-micelle adsorption of nonionic surfactants onto cement particles primarily occurs due to weak intermolecular forces. Consequently, the adsorption of nonionic surfactants to cement particles was assumed to be negligible and not a significant factor in their behavior.^{26,27} Despite their low adsorption affinity, nonionic surfactants can still induce partial hydrophobization of cement particles. Therefore, their hydrophobizing effect was incorporated into the model using contact angle measurements. Another key consideration pertained to the active agent concentrations of the surfactants. Most of the

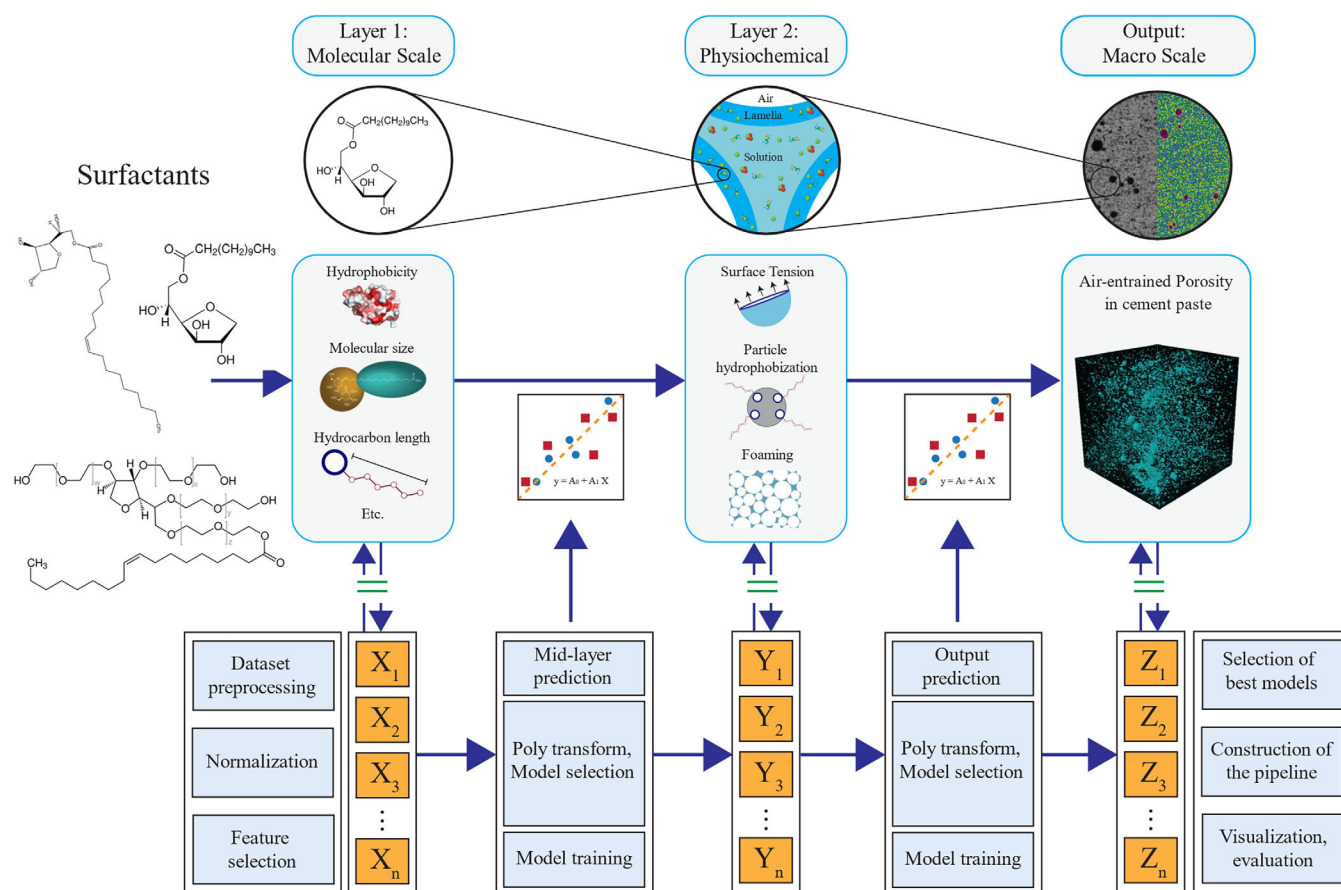


FIGURE 1 Multilayer machine learning scheme used in this study.

nonionic surfactants examined had active agent concentrations approaching 99% and were therefore assumed to be chemically pure. However, for a subset of surfactants with lower active agent concentrations, the air-entrained porosity in Layer 3 was adjusted accordingly.

3 | MATERIALS AND EXPERIMENTS

3.1 | Materials and sample preparation

A total number of 59 commercial nonionic surfactants were used in the experiments. Eighteen of these products were purchased from MilliporeSigma, and the rest were gifted by commercial chemical companies and distributors, including KLK Kolb, STEPAN, EMCO Chemical Distributors (BASF products), and Sea Land Chemicals (Evonik, KAO, and Ingevity products). The complete list of these chemicals and their molecular properties are presented in Table 1.

The chemical composition and physical properties of the Type IL cement used in this study are presented in Table 2. The cement was acquired from Titan America's plant in

Medley, Florida. Cement paste samples were cast with a 0.35 water-to-cement ratio (W/C) and a constant surfactant concentration of 0.1 wt% cement, using the mixing procedure of ASTM C305-20, but with a mixing time of 2 min at the end to achieve a better paste consistency. The paste was then vibrated for 30 s and placed in $25 \times 25 \times 300$ mm³ prism molds. The molds were covered with double layers of plastic films for the first 24 h and then cured in sealed containers for 28 days. After 28 days, they were cut into 25 mm cubes, submerged in acetone for 8 h, and then dried completely at 80°C for 24 h.

It is important to reiterate that in this study, all the components of cement paste and mixing procedures were kept constant between specimens, and the only variable parameters were the molecular properties of the surfactants. Since cement paste components such as the presence of supplementary cementitious materials, different admixtures, concentrations, W/C, and many others can affect the air-entraining performance, they have to be studied in a separate project. This is why no mixture features such as W/C or cement type are included in the model, as the model just considers the molecular properties of the surfactants for a specific cement system.

TABLE 1 Commercial nonionic surfactants used in this study.

Product information			Molecular properties						
Supplier	Product	Description	AA (%)	C#	Mw (g/mol)	HLB	EO	Density (g/cm ³)	CP (°C)
Kolb	1	Greenbentin-LM/010	Natural C12–C14 alcohol with 1 mol EO	99.9	13	244.05	3.7	1	0.86
	2	Greenbentin-LM/070 (MB)	Natural alcohol C12–14 with 7 mol EO	99.5	13	508.35	12.3	7	54.5
	3	Hedipin-CFA/100(1)	Cocos fatty acid with 10 mol EO	99.5	12	650.50	13.5	10	1.04
	4	Hedipin-PO/045 VL	Fatty acid ethoxylates, natural, linear C chain	99.5	18	500.25	8.5	5	0.97
	5	Hedipin-PS/400 G (MB)	Fatty acid ethoxylates, natural, linear C chain	99.0	18	2047.00	17.3	40	1.06
	6	Hedipin-R/2000 G	Castor oil with 200 mol EO	99.0	55	9740.00	18.0	200	1.07
	7	Hedipin-R/400 H	Castor oil with approx. 40 mol EO	99.5	55	2692.00	13.0	40	1.03
	8	Hedipin-RH/400	Hydrogenated castor oil ethoxylates, natural	97.0	55	2692.00	14.0	40	1.04
	9	Imbentin-AG/124S/110 90% (MB)	Natural alcohol C12–14 with 11 mol EO	90.0	13	684.55	14.0	11	1.00
	10	Imbentin-AG/168S/110	Natural alcohol C16–18 with 11 mol EO	99.5	17	744.55	13.0	11	0.97
	11	Imbentin-AG/168S/250	Natural alcohol C16–18 with 25 mol EO	99.5	17	1361.25	16.1	25	1.03
	12	Imbentin-AG/168S/500-AA	Natural alcohol C16–18 with 50 mol EO	99.5	17	2462.50	17.9	50	1.06
	13	Imbentin-AG/168S/800 (1)	Natural alcohol C16–18 with 80 mol EO	99.0	17	3784.00	20.0	80	1.10
	14	Imbentin-C/91/025	C9–C11 alcohol with 2.5 mol EO	99.8	10	270.13	8.1	3	0.93
	15	Imbentin-C/91/060	Oxo alcohol C9–11 with 6 mol EO	99.5	10	424.30	12.5	6	0.98
	16	Imbentin-T/400	Fatty alcohol ethoxylates, synthetic, branched C chain	99.0	13	1962.00	18.0	40	1.03
	17	Imbentin-T/65	Branched oxo alcohol C13 with 9.5 mol EO	99.5	13	640.50	13.6	10	0.99
Millipore Sigma	18	31R1	EO/PO block copolymer	0	0	3300.00	4.5	8	1.02
	19	L-31	EO/PO block copolymer	0	0	1100.00	4.0	8	1.02
	20	L-35	EO/PO block copolymer	0	0	1900.00	21.5	20	1.06
	21	F-68	EO/PO block copolymer	0	0	8400.00	24.0	164	1.06
	22	F-108	EO/PO block copolymer	0	0	14 600.00	24.0	272	100.0
	23	F-127	EO/PO block copolymer	0	0	12 600.00	21.5	190	100.0
	24	Ethylene glycol		50.0	0	62.00	8.3	1	1.11
	25	Triton X-100	<i>t</i> -Octylphenoxypolyethoxyethanol	64.0	8	625.00	13.5	10	1.07
	26	Tween 20	Polyethylene glycol sorbitan monolaurate	70.0	12	1228.00	16.7	20	1.10
	27	Tween 80	Polyoxyethylene (20) sorbitan monooleate	50.0	18	1310.00	15.0	20	1.08
	28	Span 20	Sorbitan laurate, Sorbitan monolaurate	99.0	9	346.00	8.6		1.06
	29	Span 80	Sorbitan monooleate	100.0	17	428.00	4.3		0.99

(Continues)

TABLE 1 (Continued)

Product information			Molecular properties						
Supplier	Product	Description	AA (%)	C#	Mw (g/mol)	HLB	EO	Density (g/cm³)	CP (°C)
EMCO (sourced from BASF)	30	Glucopon 50 G	C12C14-alkyl polyglucoside on inorganic and organic carrier	100.0	13				
	31	Glucopon 215 UP		100.0	9			1.14	
	32	Glucopon 225 DK		100.0	9			1.16	
	33	Glucopon 425 N		97.0	12			1.12	
STEPAN	34	MAKON L61	EO/PO block copolymer	100.0	0	2000.00	3.0	1.01	24.0
	35	MAKON L62	EO/PO block copolymer	100.0	0	2500.00	7.0	1.04	32.0
	36	MAKON L101	EO/PO block copolymer	100.0	0	3800.00	1.0	1.02	15.0
	37	MAKON TD-3	TRIDECYL ALCOHOL ETHOXYLATE, POE-3	100.0		333.00	8.0	0.93	25.0
	38	MAKON TD-30	TRIDECYL ALCOHOL ETHOXYLATE, POE-30	95.0		1496.00	17.4	0.88	100.0
	39	MAKON NF-5	Isopropanolamide-base block copolymer	100.0		1000.00	5.0	1.02	−1.0
	40	MAKON NF-12	C10–12 alcohol alkoxylate	100.0	11		6.0	0.99	15.0
	41	MAKON UD-5	C11 BRANCHED ALCOHOL ETHOXYLATE	100.0		394.00	10.9	0.98	10.0
	42	MAKON TSP-60	Polyarylphenol ethoxylates	95.0		2877.00	18.0	1.09	100.0
	43	MAKON P104	EO/PO block copolymer	99.0	0	5900.00	13.0	1.09	82.0
	44	TOXIMUL 8321	Polyalkylene oxide block copolymer	100.0		3650.00	7.0	1.03	28.0
	45	TOXIMUL 8350	Butyl-base block copolymer	100.0	0	6200.00	14.0	1.05	71.0
Millipore Sigma	46	BIO-SOFT N1-9	LINEAR ALCOHOL (C11) ETHOXYLATE, POE-9	100.0	11	569.00	13.9	0.95	78.0
	47	BIO-SOFT N91-6	LINEAR ALCOHOL (C9–11) ETHOXYLATE, POE-6	100.0	10	425.00	12.4	0.99	53.0
	48	BIO-SOFT TA-2	TALLOW AMINE ETHOXYLATE, POE-2	100.0			5.0	0.89	0.0
	49	STEP-FLOW 26F	Polyalkylene oxide block copolymer	100.0	0	7900.00	13.0	1.05	82.0
	50	TERGITOL 15-S-30	Alcohols, C12–14 secondary, ethoxylated, secondary alcohol ethoxylate 31EO	13		1520.00	17.4	1.05	100.0
	51	ECOSURF EH-6	2-Ethyl hexanol EO-PO nonionic surfactant				12.5	1.01	43.0
	52	IGEPAL CA-720	Polyoxyethylene (12) octylphenyl ether, branched	99.0	14	735.00	14.0	1.10	58.0
	53	Triton X-305 solution	Polyethylene glycol mono(4- <i>tert</i> -octylphenyl) ether	70.0	8	1526.00	17.3	1.09	100.0
	54	Pluronic 10R5	EO/PO block copolymer		0	2000.00	15.0	1.06	69.0
	55	Pluronic P-123	EO/PO block copolymer		0	5800.00	8.0	1.02	90.0
Sea-Land Chemicals (sourced from Evonik, KAO, and Ingevity)	56	Tomamine Amphoteric 16	High foaming hydrotropic surfactant	35.0					
	57	Adogen 464	Methyltrialkyl(C8–C10)ammonium chloride	85.0	9	1338.71		0.87	
	58	KAO FINDET MB-212	Alkoxylated cetyl stearyl alcohol		17		5.0		40.0
	59	Altapur Hydro P-240	C21 dicarboxylic acid derived from natural fatty acids	40.0	21			1.06	90.0

Note: Molecular properties include active agent concentration (AA), hydrocarbon chain length (C#), molecular weight (Mw), hydrophilic-lipophilic balance (HLB), ethylene oxide units (EO), density, and cloud point (CP). The surfactants were either purchased or supplied directly from manufacturers or distributors. Molecular properties of the surfactants were acquired from the manufacturer or from the literature.

TABLE 2 Mill certificate properties of cement Type IL used in the study, reported by the manufacturer Titan America.

Item	Quantity
SiO ₂	19.1
Al ₂ O ₃	4.3
Fe ₂ O ₃	3.3
CaO	63.8
MgO	0.8
Na ₂ O	0.12
K ₂ O	0.32
Alkalies	0.33
CO ₂ in cement	5.4
Sulfate as SO ₃	2.5
Limestone (%)	14.4
CaCO ₃ in limestone (%)	87
Blaine fineness (m ² /kg)	532
Loss on ignition (%)	6.5
Density (g/cm ³)	3.06

3.2 | Molecular properties (Layer 1)

The molecular properties of the surfactants were obtained from the manufacturers and used with little adjustments. Table 1 provides a comprehensive list of the chemicals used and their molecular properties. However, not all the available properties were used simultaneously, and only those that were deemed influential in the air-entraining mechanisms of surfactants in cement paste (using feature selection algorithms discussed in Section 4.1.4) were selected and used in the model development.

C# refers to the length of the hydrocarbon chain, that is, the number of saturated or unsaturated carbon atoms in the backbone of the molecule.²⁸ C# is a fundamental characteristic of surfactants and amphiphilic molecules, and in solutions, it can affect surface tension by increasing the hydrophobic interactions and disrupting the hydrogen bonding between water molecules.¹² C# can also impact the interfacial stability of the air bubbles in the solutions and properties including foaming and contact angle.^{6,28–30} Mw determines many physical and chemical interactions in the solutions, including a molecule's diffusion kinetics to air–water interfaces and interfacial activity on these interfaces.³¹ Shah et al.³¹ suggested that the lower Mw between a series of nonionic molecules, including Tween and Brij, facilitates faster adsorption at the air–water interfaces, leading to a faster equilibrium in surface tension.³¹ Hydrophilic–lipophilic balance (HLB) quantifies the extent of hydrophobicity and hydrophilicity of a surfactant molecule, with decreasing values of HLB signifying an increase in hydrophobicity.^{32,33} Hydrophobicity of the molecules can be linked to their foaming capac-

ity and stability, as well as solubility in the solutions.^{30,34} Because HLB is a metric of the overall hydrophobicity of the molecule, it is also a good indicator of the surfactant's wetting behavior on the cement surfaces and the degree to which the surfactant lowers surface tension by disrupting the hydrogen bonding network in the solution.¹² Ethylene oxide (EO) is a hydrophilic molecule, and therefore, the number of EO units in the molecule determines the hydrophilicity of the surfactant.³⁵ EO is very important in molecule's stability in the solution and in the air–water interface, and it affects the surface tension interactions and stabilizes the foams.³⁶ Cloud point (CP) is the temperature limit at which phase separation happens in nonionic surfactants.³⁷ Phase separation refers to the loss of solubility of nonionic surfactants above their CP, where the solution separates into two phases due to dehydration of EO chains at elevated temperatures, leading to the self-association of surfactants into aggregates.³⁷ This aggregate formation can reduce the number of free individual surfactants in the solution and potentially affect the surface activity at air–water interfaces and influence air entrainment. Although this study focuses on ambient temperature at 22°C, CP is nevertheless useful in describing the underlying properties of surfactant stability in the solution and the hydrophobic properties of the molecule.^{35–37} Although EO, HLB, and CP are all related to hydrophobicity, they describe different aspects of surfactant behavior. EO provides a structural measure of hydrophilicity, HLB reflects overall polarity balance, and CP serves as a measure of solubility under temperature. All three were retained in the training phase to capture a broader set of information about the surfactants' interactions in the solution.

3.3 | Physicochemical characterization (Layer 2)

Physicochemical experiments, including surface tension and foaming volume, were performed in the synthetic pore solution (SPS) adapted from Tunstall et al.¹⁰ by mixing 0.1062 M KOH, 0.0489 M Na₂SO₄, 0.037 M K₂SO₄, and 0.0212 M Ca(OH)₂ in deionized water (DI). More information on SPS preparation can be found in a previous study by the authors, where the pH of SPS was measured to be around 13.6.¹¹ Contact angle measurements and the profilometry were performed on hardened cement paste containing the surfactants.

3.3.1 | Surface tension

Surface tension is widely recognized as a critical factor in determining the performance of AEAs in concrete.^{6,38–40}

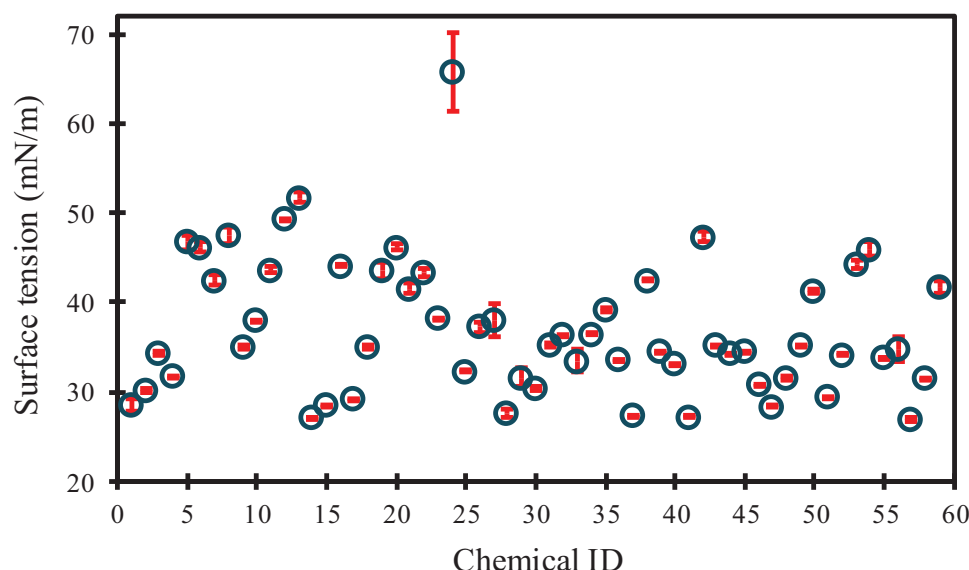


FIGURE 2 Surface tension of the surfactants of the dataset measured in synthetic pore solution (SPS) at a concentration of 0.1%.

Reducing surface tension lowers the Gibbs free energy of the solution, creating a thermodynamically favorable environment for the formation of bubbles and their uniform dispersion within the fresh concrete.⁶ Additionally, a lower surface tension decreases the destabilizing forces in the solution, such as coalescence and drainage, enabling the formation of smaller and more stable bubbles within the foam.^{6,40,41} In the highly alkaline cement pore solution, surface tension differs from that in DI water due to interactions with dissolved ions and hydration products.¹¹ Interactions among surfactants, ions, and particles can result in variable surface tension measurements, as the ability of surfactants to maintain low surface tension over time becomes important.^{5,11}

Equilibrium bulk surface tension of nonionic surfactants was measured in SPS with a BZY 201 tensiometer using a platinum plate. The equilibrium surface tension was usually achieved after a couple minutes, and three to five repetitions were performed for each surfactant.

Figure 2 displays the surface tension of the surfactants, with a concentration of 0.1% in SPS. Surface tension ranges between 27.1 and 51.7 mN/m for most surfactants in this study, with an average standard deviation of 0.4 mN/m among replicates. The only exception is ethylene glycol, which shows a high surface tension in SPS, which is similar to its typical values in DI, and has a standard deviation of 4.5 mN/m among replicates.⁴² This large range of surface tension ensures that the learning algorithm can pick up the patterns in microstructural properties associated with surface tension.

3.3.2 | Foaming behavior

The amount of foam that surfactants initially produce—referred to as foam capacity—and their stability over time—referred to as foam stability, which is the ratio of foam volume after a certain time to the initial foam volume—are very important in their performance as AEA.^{6,40} The foam volume is dictated by the molecular structure of the surfactant, which affects its ability to reduce surface tension and generate stable air bubbles in the alkaline environment of cement paste.⁶ If the foam is not stable, bubbles may coalesce or collapse before the concrete sets, resulting in a reduction in air-entrained porosity.^{40,43} Surfactants also affect foam stability through their interactions in the bubble interface.⁶ The presence of surfactants in the air–water interface of foams can affect the viscoelasticity of the interfacial layer and influence the response of these foams to deformation and instability.⁴⁴ This viscoelastic interfacial layer determines the performance of foam to external forces, including those exerted by mixing and placing of the concrete, which can destabilize weaker bubbles.⁴⁰ In the interfacial layer, surfactant molecules are oriented with their hydrophobic parts toward the air, whereas their hydrophilic heads interact with the solution.^{6,40} This arrangement forms a network of forces in the interfacial layer that gives the layer its viscoelastic properties and enhances the interfacial strength of the bubble.^{6,24,45,46} Apart from the foam capacity and stability, the size and distribution of air bubbles in foams are also dependent on the surfactant molecu-

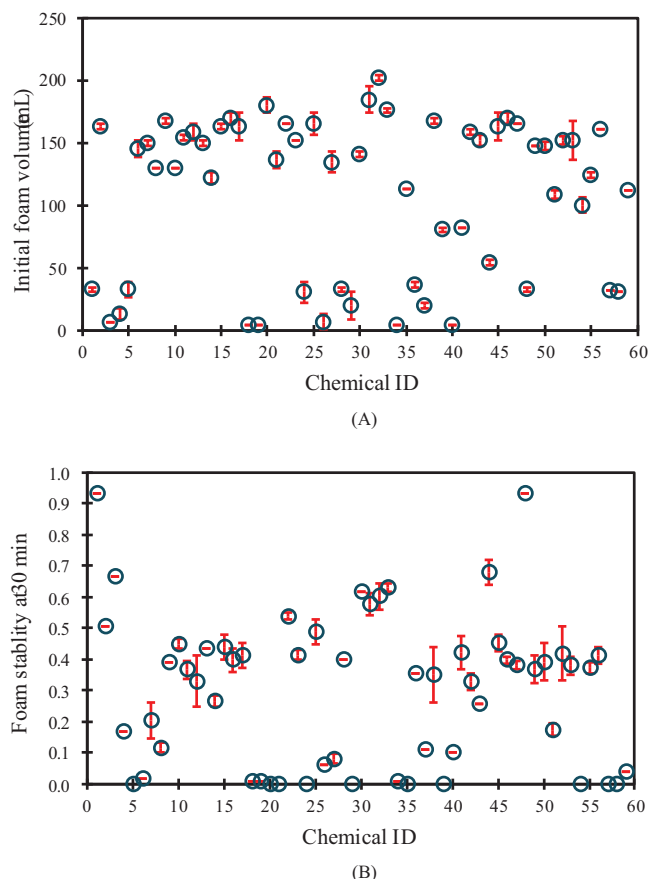


FIGURE 3 Foaming behavior of the dataset, measured in synthetic pore solution (SPS) at a concentration of 0.1%: (A) foam capacity and (B) foam stability after 30 min.

lar properties and the viscoelastic interfacial layer that they create and the degree to which this layer can limit coalescence and gas diffusion between bubbles, thereby preserving bubble size and distribution.^{30,40,43}

The blender method^{47,48} was used to measure the foaming behavior of nonionic surfactants in SPS. Surfactants with a 0.1% concentration were dissolved in SPS and mixed at low speed for 10 min. Then, 100 mL of the solution was transferred into a cylindrical blender and agitated at very high shear for 10 s, after which it was left undisturbed for up to 30 min, and the foam volume was recorded at different times. The blender has a capacity of 530 mL, an approximate radius of 3.75 cm, and a height of 12.5 cm, and it is equipped with a 6-point blade. The foaming stability, which is the ratio of foam volume after 30 min to the initial foam volume, was thus calculated.

The foaming behavior in terms of the initial foaming capacity and the stability after 30 min of the surfactants in SPS is shown in Figure 3A,B. From Figure 3A it is seen that not all surfactants are able to create foam, and some have a foam capacity close to zero, whereas some others create a large volume of initial foam. A wide distribution

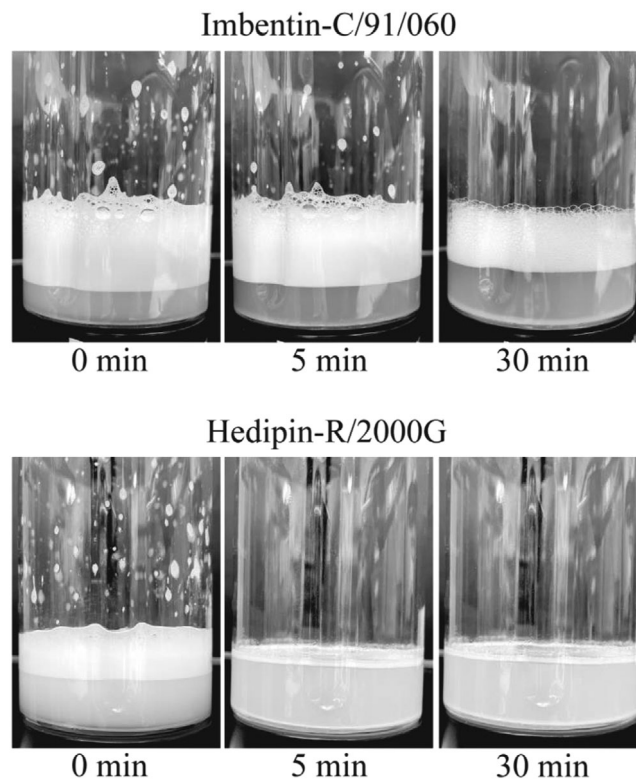


FIGURE 4 Foaming volume over time between Imbentin-C/91/060 (Chemical ID #15) and Hedipin-R/2000G (Chemical ID #6).

of foaming volume can be seen across this range. The deviation between the repetitive measurements is also small in most surfactants. Despite this, initial foaming is not enough to describe the foaming behavior of surfactants, as many of them lose a large portion of their volume in a matter of minutes. This is displayed in Figure 3B, which shows foam stability after 30 min (the ratio of the foam volume at the 30-min mark to the initial foam volume). As seen in Figure 3B, most surfactants have a moderate degree of stability, mostly concentrated around the 0.3–0.7 range. Most of the rest are located at the lower end of the stability spectrum, with values lower than 0.1.

Although both these graphs give important information about the foaming behavior of surfactants in SPS, the full picture can only be obtained when both parameters are considered. For instance, Imbentin-C/91/060 (Chemical ID #15) and Hedipin-R/2000G (Chemical ID #6) both have similar foaming capacity at around 150 mL. Although Imbentin-C/91/060 (Chemical ID #15) is able to maintain 44% of that initial volume after 30 min, Hedipin-R/2000G (Chemical ID #6) loses all its foam and drops to 1% after 30 min. This different behavior between the two surfactants is shown in Figure 4 as a factor of time. For this reason, it is important to include both foam capacity and foam stability in the machine learning model.

3.3.3 | Contact angle for particle stabilization of bubble interface

In this study, contact angle measurements have been employed to obtain a measure of cement particle hydrophobization and, subsequently, the influence of hydrophobized cement particles on the bubble interface and their effect on the air-entrained microstructure of cement paste. This happens because the surfactants in the paste can adsorb on the cement particles and hydration products and hydrophobize the paste—although to a small degree in nonionic surfactants, due to the lack of electrostatic charge.^{6,49} This increases the contact angle depending on the molecular properties and hydrophobicity of the adsorbent.⁵ This higher hydrophobicity in turn helps particles to adsorb on the interface of bubbles, increase the stability of the interface to destabilizing effects, and in some cases create a protective shell of particles around the interface.^{38,49,50} Contact angles lower than 90° indicate a hydrophilic surface, whereas angles higher than 90° are associated with a hydrophobic one.^{5,51} Plain cement paste has a very hydrophilic matrix, and water forms contact angles of typically lower than 40° on cement paste.^{5,52,53}

Pickering stabilization occurs when solid particles adsorb at an air–water interface and improve the interfacial barrier that stabilizes bubbles against coalescence.^{6,49} For particles to effectively stabilize an interface, they must possess a specific balance of hydrophilicity and hydrophobicity, often achieved through the adsorption of surfactants on the particles.⁵⁴ Because cement paste particles are highly hydrophilic (evident from their very low contact angles⁵), they do not readily adsorb at the air–water interface. However, when surfactants adsorb onto cement particles, they can hydrophobize the surface.^{5,49} This shift in hydrophobicity enables the particles to move to the air–water interface and enhance the interfacial film and subsequently the properties of the foam.^{49,54} Based on the theoretical work by Kaptay,⁵⁵ Pickering stabilization of foams is optimal at certain intermediate contact angles (around 70°–86°) at which the particles are thermodynamically favored to remain at the interface and minimize the free energy of the system.⁵⁴ Nonionic surfactants lack the charge characteristic of ionic surfactants, and as a result, the extent of adsorption onto hydrophilic surfaces is expected to be minimal.^{26,27} Instead, nonionic surfactants typically form a bilayer of adsorbed molecules or aggregate on the hydrophilic surface, with the hydrophilic head groups oriented toward the adsorbent.⁵⁶ The nature of the adsorption is primarily influenced by the concentration of the surfactant and the size of the head groups, along with other molecular properties.⁵⁶

In this study, contact angle measurements have been performed on the hardened cement paste surfaces to mea-

sure the particle hydrophobization of the matrix by the surfactants, which could contribute to the Pickering stabilization of surfactants. It is important to acknowledge that the hydrophobicity of the bulk cement matrix and the particles at the bubble interface may differ. However, directly measuring the hydrophobicity of individual particles at the bubble interface in the cement paste system is extremely challenging due to their small size and dynamic nature within the cement paste. Therefore, contact angle measurements on bulk surfaces were utilized, which offer a practical approximation of the cement particle hydrophobization, which contributes to the Pickering stabilization at bubble interfaces. The 25-mm paste cube samples were cured for 28 days and then cut in half so that the measurements could be done on their inner surfaces. Samples were dried for 24 h at 80°C and then polished with silicon carbide sandpapers of 180, 320, 600, and 1200 grits. Attension Theta Flex was used for the static advancing contact angle measurements. Droplets with a volume of 2.5 µL were placed on the polished surfaces, and the droplet profile was captured at a rate of 8.3 frames per second. The droplet profiles were then analyzed using the Young–Laplace equation in the OneAttension software to find the nominal angles between the surface and the droplet. Repetitions were performed on 3 cube samples, 6 total surfaces, and around 20–30 droplets.

Nominal contact angle can be influenced by both the hydrophobicity and the topology of the surface.^{57–59} Given the variability in surface roughness caused by air voids, contact angle measurements simultaneously reflect both hydrophobicity and topographical influences on wettability. If the contact angle on an ideally smooth surface is called the real contact angle, the effect of surface roughness in hydrophilic surfaces (<90°) will be the reduction of the angle relative to the nominal contact angle.⁵ To decouple the effect of surface roughness from contact angle measurements, surface profilometry was performed on the same surfaces using a Nanovea noncontact optical profilometer, with a 3500 µm chromatic confocal lens. Scanning was performed on areas of 5 × 5 mm² with a resolution of 5 µm. The profiles were then analyzed with the Gwyddion 2.63 software to calculate the Wenzel's roughness parameter r , which is the ratio of real rough surface to the projected smooth surface. Finally, the corrected contact angle was calculated using Wenzel's equation:

$$\cos \theta_{\text{nominal}} = r \cos \theta_{\text{Corrected}} \quad (1)$$

Figure 5 displays the contact angle measurements on cement pastes containing different surfactants. The measurements obtained for control (Chemical ID 0) are compatible with earlier studies on the same cement.⁵ In surfaces with contact angles lower than 90°, surface roughness can lower the nominal contact angle of the surface.⁵

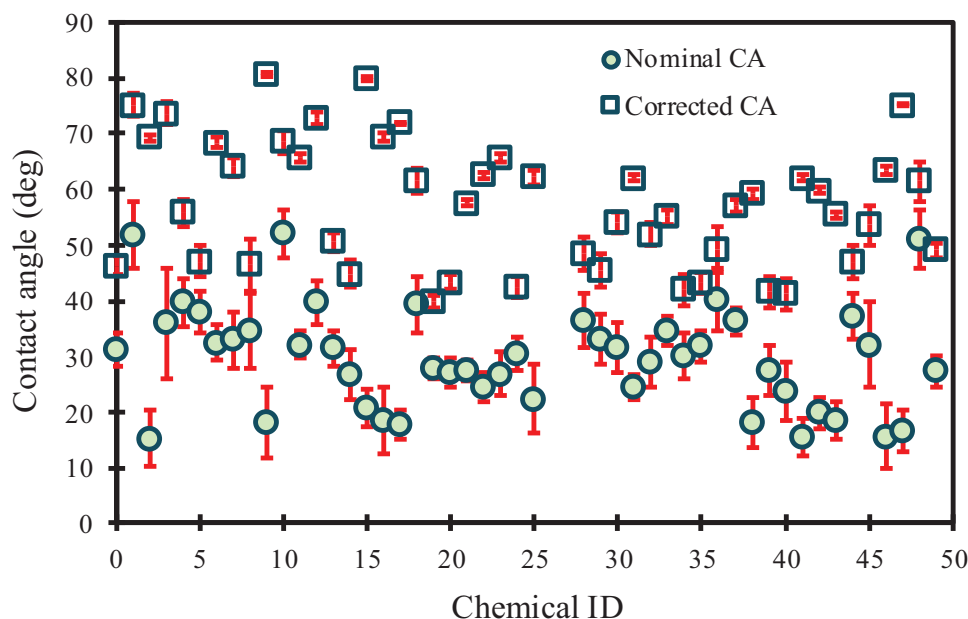


FIGURE 5 Nominal and Wenzel's corrected contact angle, measured on cement paste surface containing 0.1% surfactants. Chemical ID 0 corresponds to control cement paste.

which is seen for pastes containing surfactants. As such, to observe the hydrophobizing effect of surfactants, it is better to look at the corrected CA, which shows generally higher contact angles as high as 80° compared to control. On the other hand, not all surfactants hydrophobized the paste, and some have similar contact angles to the control. The authors have previously investigated the effect of contact angle correction and its correlation with the air-entrained properties of the cement paste.⁵ It is important to note that correcting the contact angle value introduces a certain level of bias into the measurements (especially because roughness correlates with air-entrained porosity⁵).

3.4 | Air-entrained void characterization (output layer)

The air-entrained microstructure of cement paste samples was obtained with Bruker SkyScan 1273 micro-computed tomography (micro-CT) equipment and analyzed using CTAnalyzer software. Micro-CT is capable of capturing voxels as small as $5 \mu\text{m}^3$, and as such can very accurately determine the size and location of each individual air void in the hardened cement paste and reconstruct a 3D model of the microstructure. Further numerical analyses were performed on this 3D model to determine total air-entrained porosity, void size, and void separation between air voids. The void separation is slightly different from the conventional spacing factor used in studying air voids in concrete. Although the spacing factor is a theoretical distance calculated by an extension of Darcy's law to estimate

the hydraulic pressure on the microstructure,^{60,61} the separation is the average distance between all adjacent air voids in the 3D microstructure and is calculated directly through a sphere-fitting algorithm.⁵ Cement paste samples were cut into 15 mm cubes and scanned at a pixel size of $12 \mu\text{m}$ with a voltage and an amperage of 120 kV and 125 μA , respectively. A total number of 10 projections were collected at 0.6° increments with an exposure time of 1265 ms using a 0.5 mm copper filter. The projections were then reconstructed using a 10% beam hardening correction in the dynamic range of 0–0.015.

The microstructural properties of surfactants in cement paste are displayed in Figure 6A,B, where Figure 6A shows the air-entrained porosity, and Figure 6B shows the void size and separation. Control has an air-entrained porosity close to 1%, and the nonionic surfactants increased the air-entrained porosity to as high as 25%. Void size and separation generally decreased with the addition of surfactants, with size ranging from 0.193 to 0.566 mm and separation ranging from 0.248 to 1.597 mm. Ultimately, air-entrained porosity values are used in this study to construct the machine learning model.

3.5 | Discussion on the experimental results

Figure 7A–C illustrates the trends between the experimental physicochemical properties and the air-entrained porosity observed in hardened cement paste. A clear linear relationship between surface tension and air-entrained

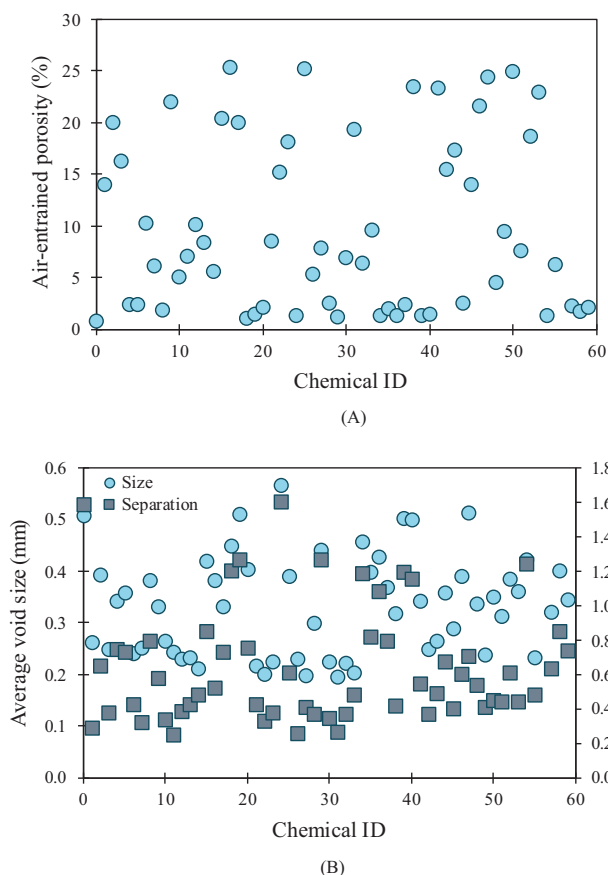


FIGURE 6 Properties of air-entrained hardened cement pastes including (A) air-entrained porosity and (B) average void size and separation, with 0.1% concentration of surfactants. Chemical ID 0 corresponds to control cement paste.

porosity cannot be observed in Figure 7A. Figure 7B shows the moderate-to-weak trend ($R^2 = 0.36$) between the 30-min foaming volume and air-entrained porosity. As will be shown in Section 5, although initial foaming volume has a strong correlation with air-entrained porosity, this relationship decays over time and is not as strong for foam volume after 30 min. The last physicochemical property is the contact angle, which has been displayed as the corrected contact angle in Figure 7C. An important note is that due to the correction of nominal contact angle with Wenzel's r , the corrected contact angle is slightly biased toward the air-entraining porosity, and it was used in the machine learning model with this in mind.

4 | COMPUTATIONAL APPROACH

4.1 | Dataset preprocessing

All preprocessing and model development was performed with the Python 3.9 programming language distribution, utilizing many open-source libraries and packages, including

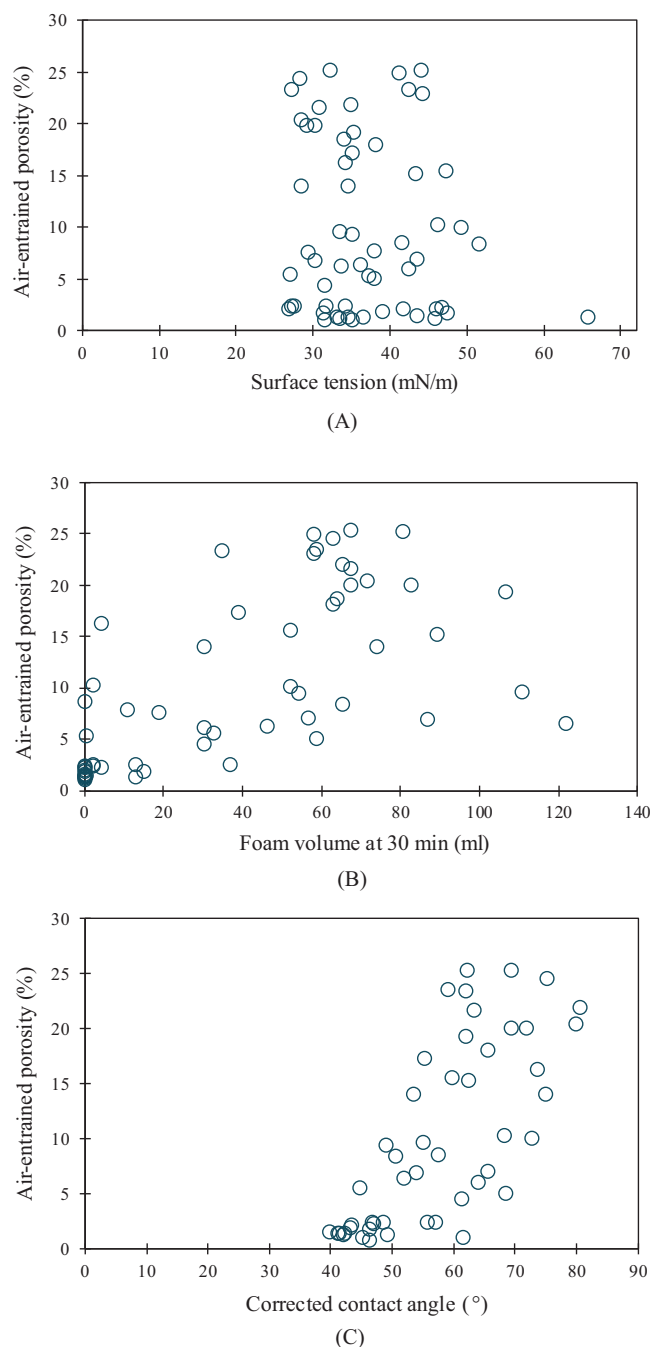


FIGURE 7 Trends between physicochemical properties of surfactants including (A) surface tension, (B) foam volume at 30 minutes, and (C) contact angle in synthetic pore solution (SPS) and the air-entrained porosity in hardened cement paste.

scikit-learn, NumPy, and pandas. The dataset was partitioned into training (70%), testing (20%), and validation (10%) subsets to ensure robust model evaluation and generalizability. The training and testing subsets were shuffled consistently and used during the model training phase and in K-fold cross-validation. On the other hand, the validation dataset was kept completely separate from the training and testing datasets and was never used during the data preprocessing or in the training phase of the model. This

is important to keep the performance of the final model unaffected by the training bias.

4.1.1 | Data imputation and normalization

Not all the molecular properties in Layer 1 were available or publicly accessible to the authors. Therefore, univariate mean imputation was performed on the available data in each column of Layer 1 to estimate the missing values. The data in Layers 2 and 3 were exclusively measured by the authors, and therefore, limited imputation was needed for them. Normalization was performed on the dataset using the z-score method so that the features have a mean of 0 and a standard deviation of 1. Normalization mitigates the impact of scale discrepancies among features, enhancing model performance and expediting convergence during training.^{62–64}

4.1.2 | Outlier detection

Outlier detection was performed on all features to identify data points that deviate significantly from the rest of the dataset. A contamination ratio of 10% was applied, using four widely employed outlier detection algorithms: elliptic envelope, local outlier factor, one-class SVM, and isolation forest. The algorithms were applied independently to all features in Layer 1 (i.e., molecular properties, such as C# and Mw) and Layer 2 (i.e., physicochemical properties of the pore solution, such as foaming, surface tension, and particle hydrophobicity). In Figure 8A,B, outlier points detected by the different algorithms are represented by distinct colors, and the radius of the circles corresponds to the anomaly score of each point relative to the rest of the data. Larger circles indicate a higher anomaly score for a specific algorithm. Because the algorithms utilize different metrics to assess outliers (e.g., Mahalanobis distance for elliptic envelope, local reachability density for local outlier factor, signed distance to the separating hyperplane for one-class SVM, and number of tree splits for isolation forest),⁶⁴ the anomaly score is normalized across each algorithm to allow for comparison between them. The outlier graphs are plotted against the first two principal components. Principal components were calculated with principal component analysis using the standardized values of all input features of Layers 1 and 2, which were used in model training. The resulting principal components are uncorrelated variables of the original features, which incorporate a large portion of feature information from the dataset. These components help visualize how surfactants cluster in chemical property space and reveal underlying categories that affect foaming and air-entrainment characteristics.

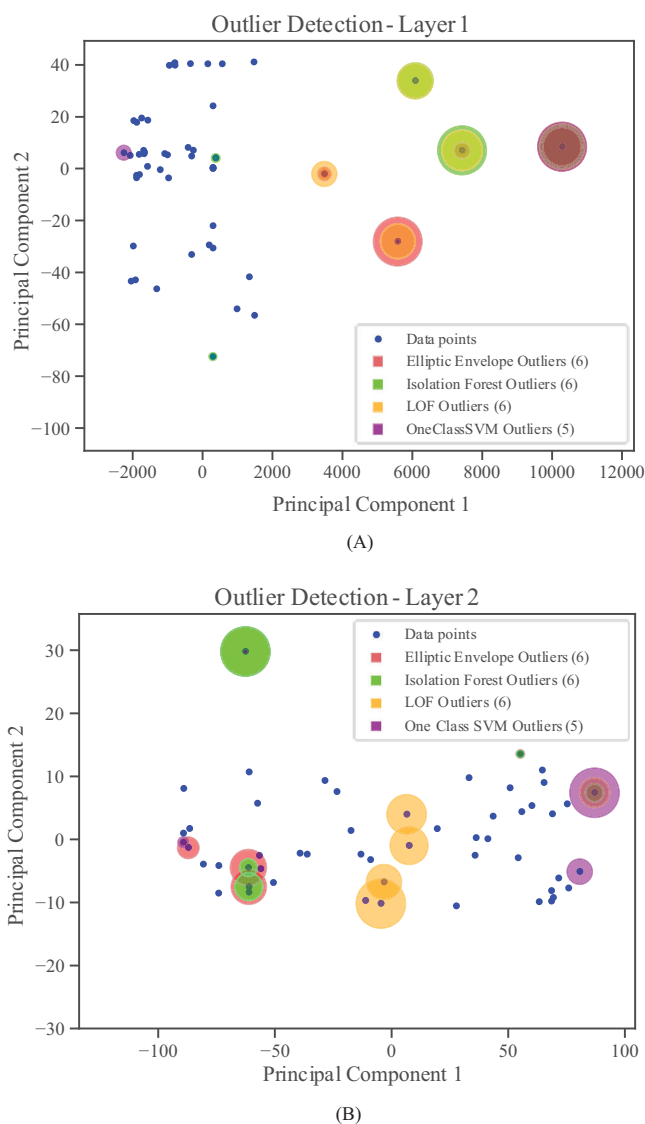
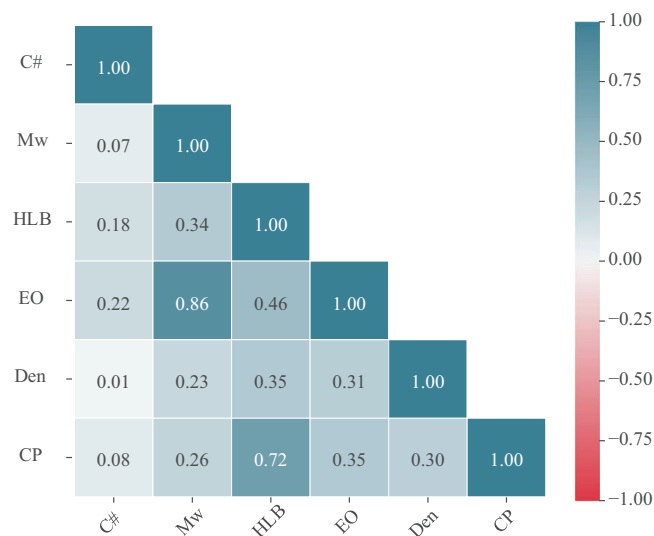


FIGURE 8 Outlier graphs for (A) Layer 1 and (B) Layer 2. Points with circles around them possess the 10% highest anomaly scores. The anomaly score for each point is represented by the radius of the circle around it.

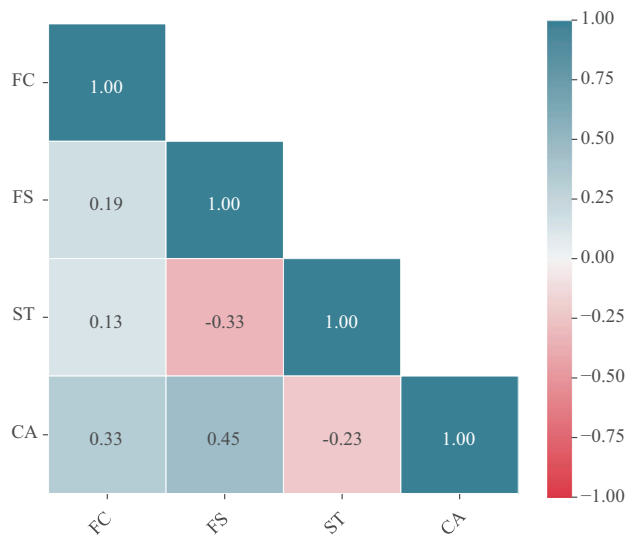
In Figure 8A, the anomalies of Layer 1 are the few data points with large Mw. These surfactants, including Hedipin-R/2000 G and F-108, possess many EO groups that make their Mw much larger than the average Mw of 2676 g/mol in the dataset. As such, the detected outliers are part of the data distribution and not incorrect data. On the other hand, the outlier graph of Layer 2 shown in Figure 8B does not display significantly dissimilar data, and the detected anomalies are consistent with the overall population. The observed outliers reflect inherent variations in surfactant molecular architecture rather than measurement inaccuracies, reinforcing the model's ability to capture diverse chemical behaviors. Therefore, no modification was performed on the detected outliers.

4.1.3 | Feature similarity

Figure 9A,B shows the pairwise correlation matrices for Layers 1 and 2 features, where the Pearson's correlation coefficient is displayed inside the correlation matrix element. Pearson's correlation coefficient measures the linear correlation between two sets of data, calculated as the



(A)



(B)

FIGURE 9 Pairwise correlation matrix for (A) Layer 1 and (B) Layer 2. The value inside the elements shows the Pearson's correlation coefficient, which indicates the linear correlation between two features. Parameters in the figures include hydrocarbon chain length (C#), molecular weight (Mw), hydrophilic-lipophilic balance (HLB), ethylene oxide units (EO), density (Den), cloud point (CP), initial foaming volume (FC), foaming stability after 30 min (FS), surface tension (ST), and corrected contact angle (CA).

ratio of their covariance to the product of their standard deviations, resulting in a value between -1 and 1 . The correlation matrix is important to detect and manage collinearity between features in the dataset, because it can negatively impact model performance.^{63,64} In this work, we have chosen Pearson's correlation coefficients higher than 0.7 as evidence of highly collinear features. Looking at Figure 9A, a high collinearity between Mw and EO is observed, which, as discussed in the previous section, is expected due to the high EO units. Another case of high collinearity exists between HLB and CP, where hydrophobicity plays a strong role in both features. Other features have much lower Pearson's coefficients. To address collinearity and prevent data redundancy during the feature selection and model training, only one of each pair of collinear features was selected in the models. Because the primary objective of the correlation matrix is to detect collinear features for model robustness and not to infer statistically significant relationships, statistical significance testing of correlation coefficients was not utilized.

4.1.4 | Feature importance and selection

To identify the most informative features in Layer 1 for predicting the properties of Layer 2, multiple feature selection algorithms were implemented. Feature importance based on univariate analysis was evaluated using three distinct metrics: the Pearson's correlation coefficient, the p -value from the F -test, and mutual information. Although Pearson's correlation coefficient and p -values quantify linear correlations, mutual information captures higher order dependencies. Additionally, feature ranking was conducted using recursive feature elimination and the coefficients of linear models. The results from all feature selection algorithms were aggregated into a single composite score for each feature, and selection was based on this consolidated value. The final set of Layer 1 features used to predict the physicochemical properties of Layer 2 is presented in Table 3, ordered by importance. These selected features were subsequently utilized for model training. As discussed in Section 4.1.3, only one feature from each pair of highly collinear variables (Mw or EO and HLB or CP) was retained, whereas the other was excluded to mitigate redundancy.

4.2 | Performance evaluation

The coefficient of determination (R^2) is a unitless metric used to assess the goodness of fit of the model on the data and can have a value ranging from 0 to 1 , with 1

TABLE 3 Ranking of the Layer 1 features, based on their importance in predicting features of Layer 2.

Predicted feature in Layer 2	Selected features in Layer 1			
	HLB	C#	Den	EO
FS	EO	Den	HLB	C#
ST	Den	CP	C#	EO
CA	HLB	C#	Mw	Den

Note: Results are obtained from feature selection.

Abbreviations: CP, cloud point; Den, density; EO, ethylene oxide; HLB, hydrophilic-lipophilic balance; Mw, molecular weight; ST, surface tension.

being an ideally perfect fit. Moreover, a combination of metrics like mean absolute error (MAE) and root mean squared error (RMSE), and techniques including cross-validation and regularization, were employed to provide a more robust evaluation of model performance and generalizability. Feature selection was performed, and the relevant features were used for each model, as was discussed in the preceding section. Additionally, model regularization was performed on the models to control overfitting of the model. Although it is important to achieve a reasonably high value for R^2 , a high R^2 by itself is not sufficient in evaluating the generalization performance of the model beyond the training dataset.¹²

To calculate the prediction error for the models, the RMSE and MAE are used, which show the error in the same unit as the original features. RMSE and MAE estimate the average error between the predicted values $\hat{y}_i$ and the ground truth y_i , in a dataset composed of n individual samples, and are defined as

$$\text{RMSE} = \sqrt{\frac{\sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2}{n}} \quad (2)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=0}^{n-1} |y_i - \hat{y}_i| \quad (3)$$

It is crucial to assess and compare the performance of the proposed model with a benchmark model. However, there are no comparable models in the literature that could be used as a benchmark, and as such, different automatic machine learning (AutoML) packages were considered. AutoML is a type of software package that can automate the process of selecting, training, and optimizing machine learning models by streamlining many repetitive processes. However, manual training provides better insight, control, and interpretability for complex systems such as the current study. This is why the models of this study were trained manually, and AutoML was only used as a benchmark. Many AutoML packages are avail-

able, from which PyCaret⁶⁵ was selected and utilized as a benchmark model to assess the performance of the developed models. The dataset was preprocessed with the same parameters as the main model, and models were developed using PyCaret's automatic model development algorithm. Finally, the model developed with PyCaret was used to evaluate the performance of this study's model on the evaluation dataset.

4.3 | Model selection

The connection between the different layers of the multilayer model is established through regularized linear models. Linear models are advantageous due to their high interpretability, making them ideal for elucidating the complex relationships between molecular, physico-chemical, and microstructural attributes.⁶⁶ The built-in regularization in the loss function of many linear models provides an effective mechanism to address overfitting, particularly in smaller datasets with a large number of features.^{63,64} Several algorithms from the scikit-learn library were employed during training to identify the most effective predictive model for the dataset. These include Lasso, Ridge, ElasticNet, SGDRegressor, HuberRegressor, PoissonRegressor, GammaRegressor, and TweedieRegressor. Lasso and Ridge are linear models with one term of regularization, L1 for Lasso and L2 for Ridge.^{63,64} The L1 and L2 terms control the model's regularization and can help prevent overfitting. ElasticNet combines both L1 and L2 regularization (with different ratios of L1/L2) to create a more balanced model, capable of dealing with highly correlated features efficiently.⁶⁴ SGDRegressor uses an iterative gradient descent optimization algorithm to minimize the loss function and find the best coefficients for the linear model.⁶⁴ It uses either L1, L2, or a combination of them to regularize the model, and it is more sensitive to the hyperparameters due to its stochastic convergence methodology.⁶⁴ HuberRegressor is a version of the least squares regression that uses a squared loss function for small values and a linear absolute loss function for large outliers.^{63,64} This hybrid loss function penalizes large residuals to a lesser degree and makes Huber regression more effective at handling outliers that might affect the coefficients significantly.⁶³ The next three models are generalized linear models that are adept at learning the datasets with specific distributions: Poisson distribution for PoissonRegressor and Gamma distribution for GammaRegressor. TweedieRegressor can handle a wide range of distributions, including compound Poisson, Gamma, and inverse Gaussian, through the power hyperparameter.⁶⁴

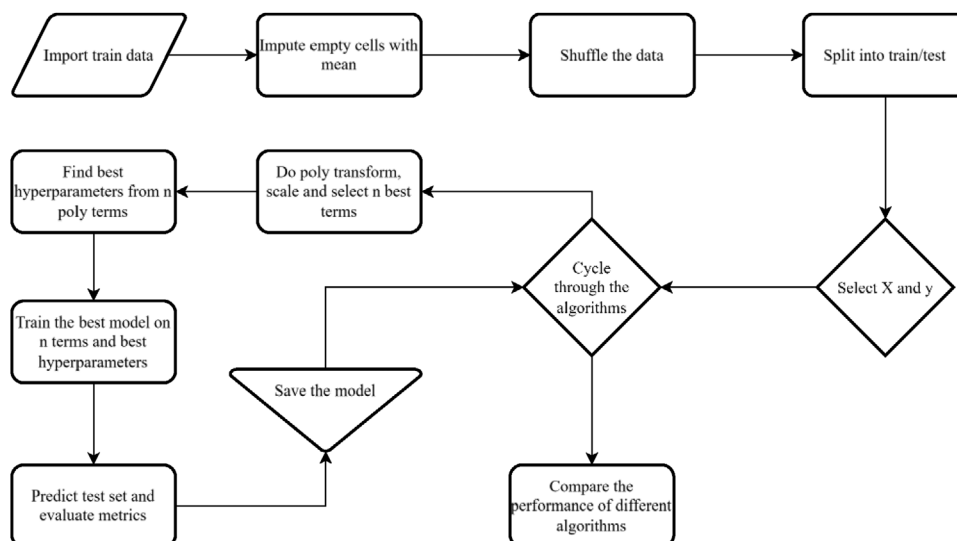


FIGURE 10 Development flowchart for each model in the hierarchical scheme.

4.4 | Model training

A custom optimization iterative algorithm is developed to find the best hyperparameters for different algorithms, as shown in Figure 10. A second-degree polynomial transform is performed on the features to find the best combinations of the features. The poly features are then ranked based on the absolute value of their coefficient. The poly features with the highest coefficients are used for training and evaluation: 4 for Layer 2 models and 5 for Layer 3 models. The regularization hyperparameter is initially determined through a preliminary training phase on the dataset. During this phase, the R^2 and RMSE metrics were assessed across thousands of incremental values of the regularization hyperparameter. The final selection of this hyperparameter was determined by optimizing the trade-off between model fit and generalization. Once the regularization hyperparameter was established, the parameter space was systematically explored to identify the optimal combination of hyperparameters that maximized performance on the test set. Models were compared to identify the most effective training algorithm and hyperparameter configuration, and the best-performing model was retained for further processing. Cross-validation was employed during training to ensure a rigorous evaluation. In cross-validation, the training dataset was partitioned into n subsets, and the model underwent n training iterations, with each iteration using one subset for validation while training on the remaining $n - 1$ subsets. After completion, the performance metrics from all iterations were averaged and reported as the final model evaluation. In this study, 5-fold cross-validation was utilized during both training and testing.

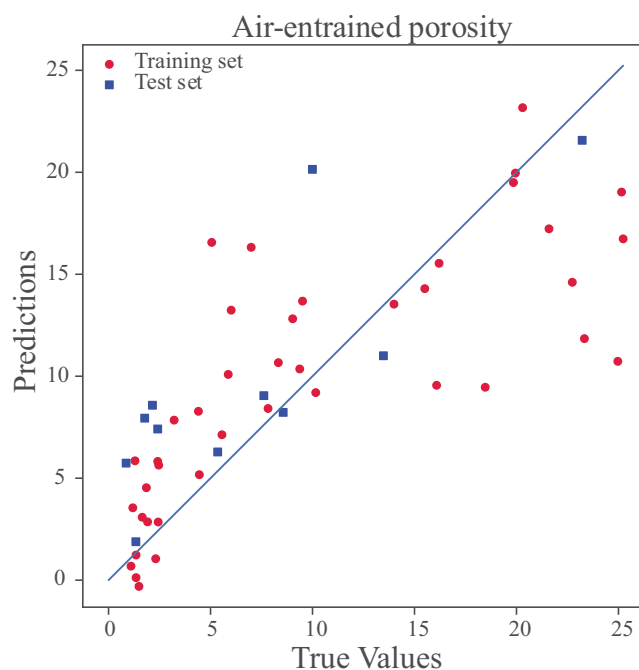


FIGURE 11 True versus predicted values on the training and test datasets for air-entrained porosity. Only the best algorithm, selected for the final model, is shown. The model has an $R^2 = 0.61$, root mean squared error (RMSE) = 4.03, and a mean absolute error (MAE) = 3.64 on the test/train dataset.

4.5 | Trained model

Figure 11 shows the performance of the final model that achieved the best result on training and test datasets. The final model is assembled from individual models that connect different layers, according to our machine learning

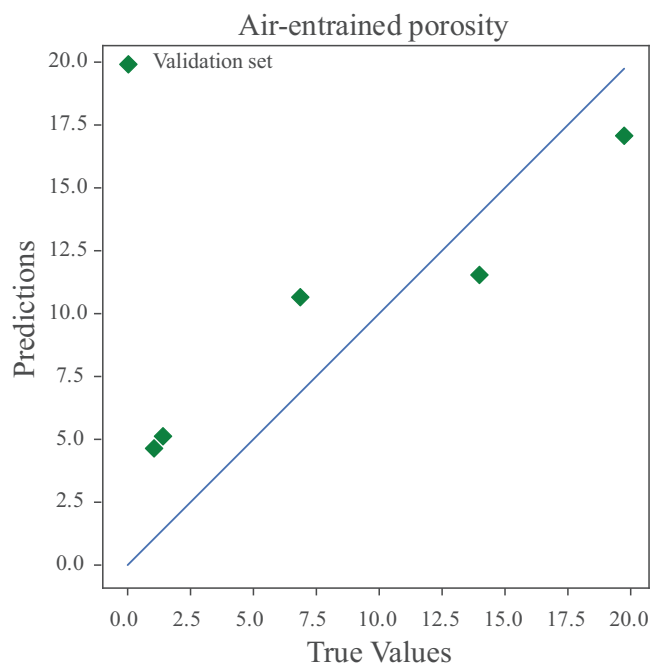


FIGURE 12 True versus predicted values on the validation dataset for air-entrained porosity. The data points display the performance on the final model.

TABLE 4 Performance of the developed model on the validation set in predicting air-entrained porosity.

	AE porosity		
	MAE	RMSE	R^2
This study	3.24	3.29	0.80
PyCaret	5.84	6.01	0.32

Note: AutoML benchmark model PyCaret is also presented for reference. Abbreviation: MAE, mean absolute error; RMSE, root mean squared error.

multilayer scheme displayed in Figure 1. In Figure 11, the predicted values of air-entrained porosity calculated by the model are plotted against the ground truth values obtained from the micro-CT analysis. The distance of the points from the identity line shows the goodness of fit of the model on the data in the training and test sets. Figure 12 displays the performance of the final assembled multilayer model on the validation dataset, and Table 4 shows the performance metrics of this study's model and that of the PyCaret benchmark model. As can be seen, this study's model is showing notably better performance metrics than PyCaret in predicting air-entrained porosity. The model predicts the validation data with an R^2 of 0.8 and an RMSE of 3.29. As discussed in Section 4.2, PyCaret is an AutoML package that automates data preprocessing and training once the dataset is inputted. Because this study is novel in its scope and implementation, and there is no comparable model in literature that can predict the air-entraining performance of nonionic surfactants in

cement paste from their molecular properties, the authors used PyCaret to assess the performance of the developed model with PyCaret. In comparison, the PyCaret benchmark model has a much lower R^2 and a higher RMSE of 6.01. Furthermore, the model developed in this study is able to predict the air-entrained porosity of nonionic surfactants consistently in all ranges of low, medium, and high air-entrained porosity. In Figures 11 and 12, it is observed that the model is systematically overestimating the air-entrained porosity in lower ranges and underestimating in higher ranges.

5 | DISCUSSION ON THE RELATIONSHIP AMONG MODEL FEATURES

Figure 13A–C shows some of the important features of Layer 1 that display clear patterns with the physicochemical features in Layer 2, including foaming capacity and surface tension. Figure 13A demonstrates the role of HLB on the foam capacity of surfactants in cement pore solution. A surfactant's ability to create foam depends to some degree on its amphiphilic structure because it determines surface tension at the air–water interface and the stability of the interfacial film.⁶⁷ As indicated in Figure 13A, an HLB value in the range of 12–25 generally resulted in the highest volume of initial foam in the pore solution. Surfactants with HLB values in this range dissolve easily in the pore solution while providing adequate surface activity through their hydrophobic regions. Although not all surfactants within this range of HLB possess high foaming capacity in the pore solution, a clear increase in foaming is observed. However, this higher HLB does not mean that surfactants in this range lack surface activity, and as is observed in Figure 13C, surfactants in all ranges of HLB are capable of reducing surface tension to below ~50 mN/m. On the other hand, surfactants with lower HLB values are more hydrophobic and generally less successful at producing initial foam in the pore solution, which in part may be due to hydrophobic micellization of the surfactants.⁶⁸ Similarly, it can be anticipated that surfactants with HLB values higher than 25—although not available in this study—will also be less effective in producing foam due to somewhat weaker hydrophobic interactions and high destabilizing forces in the interface.⁶⁹

Figure 13B illustrates the effect of Mw on the surface tension of nonionic surfactants in cement pore solution. As shown in the figure, the lowest surface tensions correspond to surfactants with the smallest Mws, and as the Mw increases, so does the surface tension of surfactants. This behavior can be attributed to some degree to the interactions associated with the hydrophilic EO units in the sur-

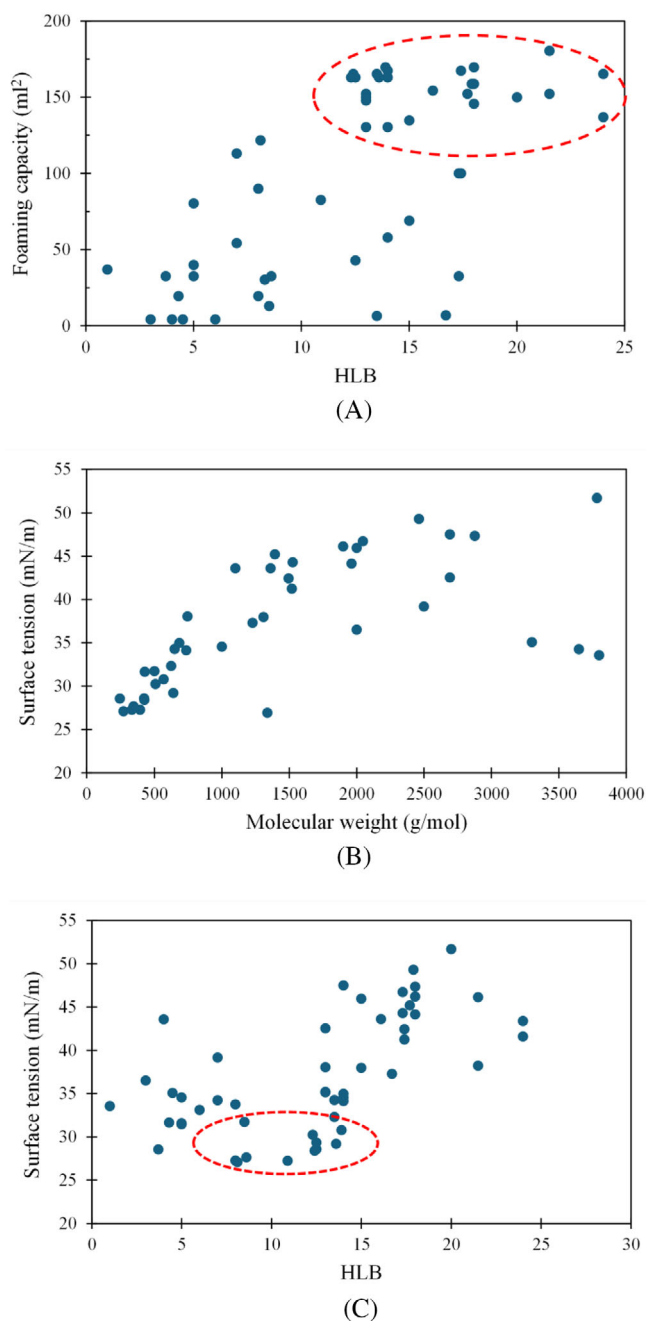


FIGURE 13 Important polynomial features for Layer 2, showing (A) HLB and foaming volume, (B) molecular weight and surface tension, and (C) HLB and surface tension. Surfactants with molecular weights above 4000 g/mol are omitted in (B). HLB, hydrophilic–lipophilic balance.

factants and their interactions at the air–water interface. As indicated in the correlation matrix (Figure 9A), Mw and EO units exhibit a strong correlation, with larger surfactants in this study typically having a higher number of EO units. Kanokkarn et al.⁷⁰ reported that as the number of EO units increases in a nonionic surfactant in a saturated solution, the surfactants tend to have a lower concentration at the air–water interface, which subsequently results in

higher surface tension. As Mw increases, some surfactants may exhibit either excessive hydrophilicity or hydrophobicity, which is respectively due to a high number of EO units or long hydrocarbon chains in their structure. Highly hydrophilic surfactants are unable to adsorb effectively at the air–water interface, thereby limiting their surface activity.^{70,71}

In contrast, excessively hydrophobic surfactants may exhibit poor solubility in water and may be prone to precipitation or micelle formation prior to adsorption at the interface.⁷¹ This is in line with previous works such as the study by Sheng et al.⁷² where it was found that optimum values of both HLB and molecular size are needed simultaneously to effectively lower the interfacial tension in oil–water interfaces. Therefore, and in the absence of more data on the subject, we believe that a balance of hydrophobic and hydrophilic interactions—hydrocarbon chain length versus the number of EO units, respectively—is required for high surface activity of the surfactants in this study, as displayed in Figure 13C for HLB. Surfactants with larger hydrocarbon chain lengths generally have stronger hydrophobic interactions, which enhance their adsorption at the air–water interfaces.^{38,70} This adsorption disrupts the hydrogen bonding between water molecules and reduces surface tension.^{6,73,74} However, it is important to note that other properties besides chain length, including hydrocarbon chain branching, chain saturation and rigidity, and the presence of an aromatic nucleus, can also affect the hydrophobicity of the hydrocarbon chain and its surface activity at the interface.^{38,75} The effect of HLB on hydrophobicity was discussed in Section 3.2. In Figure 13C, surfactants with higher hydrophobicity (HLB < 12) generally show lower surface tensions in cement pore solution compared to the rest of surfactants. However, generally the lowest surface tensions are associated with surfactants that have intermediate HLB values around 7–12, as they achieve the best balance between hydrophobicity and hydrophilicity for reducing surface tension at air–water interfaces. However, it is important to note that some surfactants with lower HLB values (e.g., HLB ~4) also showed strong surface activity. This suggests that surface tension reduction is influenced by a combination of molecular properties beyond HLB alone.

Figure 14A,B illustrates the relationship between two primary features of the developed model and the air-entrained porosity of hardened cement paste. As shown in Figure 14A, the combined effects of foaming stability and contact angle serve as an indicator of the air-entrained porosity. The foaming stability of nonionic surfactants is a key factor in determining the air-entrained porosity in hardened cement paste. Surfactants with high foaming stability generate bubbles with stable films that resist coalescence and drainage, thereby maintaining their size and shape throughout hydration.⁴⁰ As previously discussed,

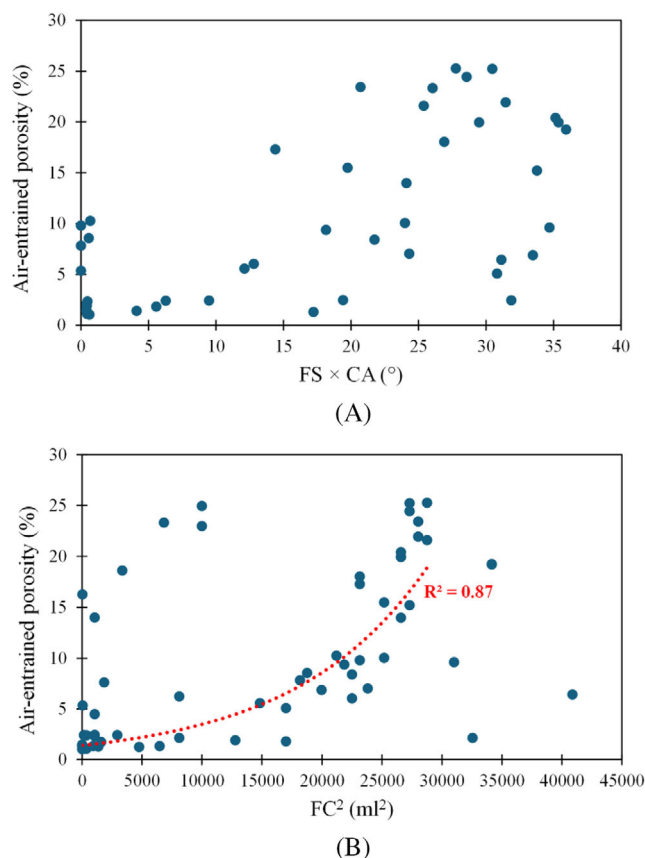


FIGURE 14 Influential polynomial features for Layer, including (A) combined effect of foam stability and contact angle on air-entrained porosity, and (B) foaming capacity on air-entrained porosity. The R^2 factor for (B) only corresponds to the upward trend, as mentioned in Section 5.

contact angle serves as an indicator of the extent of particle hydrophobization and can assess the stability of this interfacial layer via Pickering stabilization.^{54,55} At lower $FS \times CA$ values, air-entrained porosity is relatively low, indicating that low foaming stability and contact angle are insufficient to produce significant air entrainment. As $FS \times CA$ increases, there is a noticeable range of air-entrained porosity from low to very high. This indicates that although higher $FS \times CA$ is more favorable for air bubble formation and retention, there are other important factors at play as well. Another feature is the foaming capacity, which reflects the ability of a surfactant to produce bubbles during mixing, and as FC^2 , it emphasizes the exponential impact of bubble volume on air-entrained systems.⁵ Figure 14B displays the effect of FC^2 as an important feature in the model on the air-entrained porosity in hardened cement paste. Two general patterns can be distinguished in the plots: a scatter of points in FC^2 values lower than 1000 mL², and a clear upward exponential pattern starting at 0 and continuing up to 30 000 mL² (with an R^2 of 0.87).

6 | CONCLUSIONS AND FUTURE DIRECTIONS

This study demonstrates the application of a novel multi-layer machine learning model to predict the air-entrained porosity in cement pastes from the molecular characteristics of nonionic surfactants. The three-layer approach was developed to predict air-entrained porosity of the macro voids in the hardened cement paste microstructure, using molecular, physicochemical, and microstructural properties of surfactants. The integration of these molecular, physicochemical, and microstructural data in a single predictive framework significantly outperforms conventional experimental methods, setting a new benchmark in admixture design and optimization. A large series of experiments related to air-entrained porosity were performed to build a novel dataset of mechanistic parameters related to air-entraining of nonionic surfactants, which was subsequently used for model development. The results showed that foaming behavior and particle hydrophobization are significant factors influencing the air-entrained porosity of nonionic surfactants. The findings not only validate the efficacy of the proposed approach but also provide new insight into the mechanistic interplay between surfactant molecular properties and its air-entraining performance in cement paste, which is critical for improving air-entraining agents and enhancing freeze–thaw durability in concrete. This work paves the way for future research into developing surfactants as AEAs in cementitious materials. Future research can investigate and measure the physicochemical properties that pertain to air-entrained void size and separation, which are needed to develop a model predicting them. Moreover, work can be done looking at the effect of surfactant concentration, ionic surfactants, interaction with other admixtures, and the influence of other cementitious and supplementary cementitious materials in concrete.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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